

- 2 (a) Describe the purpose of a user-defined data type.

.....

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.....

.....

..... [2]

- (b) Define, using pseudocode, the following enumerated data types:

- (i) `SchoolDay` to hold data about the days students are usually in school.

.....

..... [1]

- (ii) `WeekEnd` to hold data about the days that are not school days.

.....

..... [1]

- (c) Define, using pseudocode, the composite data type `ClubMeet`. This will hold data about club members that includes:

- first name and last name
- the two days they attend:
 - one on a school day
 - one not on a school day.

Use the enumerated types you created in **part (b)**.

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..... [4]

- 8 (a) State **two** factors that may affect the performance of a sorting algorithm.

.....

.....

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..... [2]

- (b) The given algorithm is a simple bubble sort that arranges a set of scores stored in a one-dimensional array into **descending** order, and orders the corresponding students' names stored into a two-dimensional array in the same order as the scores. All the arrays are indexed from 1.

The contents of both arrays after sorting are shown.

	Score
1	98
2	97
...	
248	5
249	3

	Name	
	1	2
1	Smithfield	Tom
2	Johnson	Jane
...		
248	Peters	Jade
249	Allen	John

```

YearSize ← 249
Flag ← TRUE
WHILE Flag = TRUE
    Flag ← FALSE
    FOR Student ← 1 TO YearSize - 1
        IF Score[Student] < Score[Student + 1] THEN
            Temp1 ← Score[Student]
            Temp2 ← Name[Student,1]
            Temp3 ← Name[Student,2]
            Score[Student] ← Score[Student + 1]
            Name[Student,1] ← Name[Student + 1,1]
            Name[Student,2] ← Name[Student + 1,2]
            Score[Student + 1] ← Temp1
            Name[Student + 1,1] ← Temp2
            Name[Student + 1,2] ← Temp3
            Flag ← TRUE
        ENDIF
    NEXT Student
ENDWHILE

```

Write an algorithm, using pseudocode, that will perform the same task using an insertion sort.

[6]

- 9 (a) Describe what is meant by **an imperative (procedural)** programming language.

.....

.....

.....

..... [2]

- (b) Describe what is meant by **a declarative** programming language.

.....

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.....

..... [2]

- (c) Identify the programming paradigm for each of these program code examples.

Program code example	Programming paradigm
male(john). female(ethel). parent(john, ethel).	
FOR Counter = 1 TO 20 X = X * Counter NEXT Counter	
Start: LDD Counter INC ACC STO Counter	
public class Vehicle { private speed; public Vehicle() { speed = 0; } }	

[4]

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- 2 Draw **one** line from each programming paradigm to its **most appropriate** description.

Programming paradigm	Description
	Programs using the instruction set of a processor
Declarative	Programs based on events such as user actions or sensor outputs
Imperative	Programs using the concepts of class, inheritance, encapsulation and polymorphism
Low-level	Programs with an explicit sequence of commands that update the program state, with or without procedure calls
Object-oriented	Programs that specify the desired result rather than how to get to it

[4]

- 3 Enumerated and pointer are two non-composite data types.

- (a) Write **pseudocode** to create an enumerated type called `Parts` to include these parts sold in a computer shop:

Monitor, CPU, SSD, HDD, LaserPrinter, Keyboard, Mouse

.....

 [2]

- (b) Write **pseudocode** to create a pointer type called `SelectParts` that will reference the memory location in which the current part name is stored.

.....

 [2]

10 (a) State **three** essential features of **recursion**.

- 1
-
- 2
-
- 3
-
- [3]

(b) Explain the reasons why a stack is a suitable Abstract Data Type (ADT) to implement recursion.

-
-
-
-
-
-
-
- [3]

(c) Identify **two** ADTs other than a stack.

- 1
- 2
- [2]

1 Data types can be defined using pseudocode.

The data type, `LibraryRecord`, is defined in pseudocode as:

```
TYPE LibraryRecord
  DECLARE Title : STRING
  DECLARE Fiction : BOOLEAN
  DECLARE Author : STRING
  DECLARE NumberOfCopies : INTEGER
ENDTYPE
```

A variable, `LibraryBook`, is declared in pseudocode as:

```
DECLARE LibraryBook : LibraryRecord
```

(a) Write **pseudocode** statements to assign:

- A Level Computer Science **to** `Title` **of** `LibraryBook`
- `FALSE` **to** `Fiction` **of** `LibraryBook`.

.....

 [2]

(b) The type definition for `LibraryRecord` is changed.

- (i) The value for `NumberOfCopies` must be between 1 and 10 inclusive.

Write the updated line of **pseudocode** from the type definition of `LibraryRecord` to implement the change.

.....
 [1]

- (ii) Every copy of every book is now uniquely identified by an accession number, `AccessionNumber`, as it is added to the library. Each library record will include one or more accession numbers. Each accession number is an integer.

Write the extra line of **pseudocode** needed in the type definition of `LibraryRecord`.

.....

 [2]

- (c) A record is a user-defined composite data type.

Explain what is meant by a **user-defined composite data type**.

Include an example of **another** user-defined composite data type in your answer.

.....

.....

.....

.....

.....

.....

[3]

2 A declarative language is used to represent the following facts about cats.

```
01 type(leopard, wild).
02 type(lion, wild).
03 type(cheetah, wild).
04 type(savannah, hybrid).
05 type(persian, domestic).
06
07 hair(leopard, medium).
08 hair(lion, short).
09 hair(cheetah, medium).
10 hair(savannah, medium).
11 hair(persian, long).
12
13 spots(leopard, yes).
14 spots(lion, no).
15 spots(cheetah, yes).
16 spots(savannah, yes).
17 spots(persian, no).
```

These clauses have the following meaning:

Clause	Meaning
01	A leopard is a type of wild cat.
08	A lion has short hair.
16	A savannah has spots.

- (a) More facts are to be included. A **caracal** is a wild cat with short hair.

Write the additional clauses to record these facts.

18

19

[2]

- (b) Using the variable `Cat`, the goal:

```
hair(Cat, medium)
```

returns

```
Cat = leopard, cheetah, savannah
```

Write the result returned by the goal:

```
hair(Cat, long)
```

Cat = [1]

- (c) (i) Write the goal, using the variable `Pet`, to find all the domestic cats.

.....

..... [1]

- (ii) Write the goal, using the variable `WildSpotty`, to find all the wild cats with spots.

.....

.....

.....

..... [2]

- 7 A program is to be written using Object-Oriented Programming (OOP) for a shop that sells knitting yarn. There are three types of yarn: acrylic, wool or mix.

The following data are stored for each type.

- Name
- Colour
- Batch code
- Weight
- Number of balls of yarn in stock (can be edited)
- Type of yarn

The following statements apply to yarn.

- Acrylic can be soft or not soft.
- Wool can be lamb, merino or alpaca.
- Mix contains a percentage of acrylic.

Each type of yarn has a method that will display all the information about the yarn.

- (a) Complete this class inheritance diagram to show the **properties**, **methods** and **inheritance**.

Yarn
Name: STRING Colour: STRING BatchCode: STRING Weight: INTEGER NumberBalls: INTEGER Type: STRING
Constructor() EditNumberBalls() YarnInfo()

Acrylic
.....
Constructor()

Wool
.....
Constructor()

Mix
Percentage: INTEGER
Constructor() YarnInfo()

[5]

(b) Describe what is meant by the terms **properties**, **methods** and **inheritance**.

Properties

.....

.....

.....

Methods

.....

.....

.....

Inheritance

.....

.....

.....

[6]

1 Data types can be defined using pseudocode.

The data type, `BuildingRecord`, is defined in pseudocode as:

```
TYPE BuildingRecord
  DECLARE BuildingID : INTEGER
  DECLARE BuildingGroup : STRING
  DECLARE OwnerName : STRING
  DECLARE BuildingAddress : STRING
  DECLARE DateLastSold : DATE
  DECLARE PriceLastSold : REAL
ENDTYPE
```

A variable, `BuildingRegister`, is declared in pseudocode as:

```
DECLARE BuildingRegister : BuildingRecord
```

(a) Write **pseudocode** statements to assign:

- 1067 to the `BuildingID` of `BuildingRegister`
- house to the `BuildingGroup` of `BuildingRegister`

.....

.....

.....

..... [2]

(b) The type definition for `BuildingRecord` is changed. The data type for `BuildingGroup` is changed to an enumerated type, `BuildingType`, with values of house, bungalow, apartment and farm.

(i) Write the type declaration for `BuildingType` in **pseudocode**.

.....

.....

.....

..... [2]

(ii) Write the new declaration for `BuildingGroup` in **pseudocode**.

.....

..... [1]

(iii) Write the new **pseudocode** statement to assign house to `BuildingGroup` of `BuildingRegister`.

.....

..... [1]

- (c) The program is to be rewritten using Object-Oriented Programming (OOP). The data type `BuildingRecord` is to be changed to a class, `BuildingClass`.

The properties for `BuildingClass` are `BuildingID`, `BuildingGroup`, `OwnerName`, `BuildingAddress`, `DateLastSold` and `PriceLastSold`.

All the properties are set to `PRIVATE`, for example:

```
PRIVATE PriceLastSold : REAL
```

- (i) Write the declaration in **pseudocode** for `OwnerName` as `PRIVATE`.

.....
 [1]

- (ii) Explain why the properties have been set to `PRIVATE`.

.....

 [2]

2 A declarative language is used to represent the following facts about a school.

```

01 teaches(alan, mathematics).
02 teaches(ioana, geography).
03 teaches(nina, history).
04 teaches(alan, statistics).
05
06 studies(ahmed, history).
07 studies(freya, history).
08 studies(kim, history).
09 studies(freya, geography).
10 studies(hua, mathematics).
11 studies(hua, statistics).
12 studies(hua, geography).
13
14 tutors(alan, kim).
15 tutors(alan, hua).
16 tutors(alan, freya).
17 tutors(nina, ahmed).

```

These clauses have the following meaning:

Clause	Meaning
01	Alan teaches mathematics.
06	Ahmed studies history.
14	Alan is Kim's tutor.

(a) More facts are to be included. Sam studies history and Nina is his tutor.

Write the additional clauses to record these facts.

18

19 [2]

(b) Using the variable `Student`, the goal:

```
studies(Student, history)
```

returns

```
Student = freya, ahmed, kim
```

Write the result returned by the goal:

```
studies(Student, geography)
```

Student = [1]

- (c) Write the goal, using the variable x , to find all the students who have a tutor that teaches them. For example, Hua has Alan for a tutor and is also taught mathematics by Alan.

.....

.....

.....

..... [4]

- 3 The TCP/IP protocol suite has four layers. The application layer provides user services.

- (a) Identify **two** protocols used by this layer. Describe the use of each protocol.

Protocol 1

Description

.....

.....

Protocol 2

Description

.....

..... [4]

- (b) Identify **two other** layers of the TCP/IP protocol suite. Describe the function of each layer.

Layer 1

Description

.....

.....

Layer 2

Description

.....

..... [4]

8 A binary search or a linear search can be used to look for a specific value in an array.

(a) Complete this pseudocode algorithm for a linear search.

```

DECLARE MyList : ARRAY[0:9] OF INTEGER
DECLARE MaxIndex : INTEGER
DECLARE Index : INTEGER
DECLARE Found : BOOLEAN
DECLARE ValueToFind : .....

INPUT ValueToFind
Found ← FALSE
Index ← 0
MaxIndex ← .....

REPEAT
    IF MyList[Index] = ValueToFind THEN
        Found ← TRUE
    ENDIF
    Index ← .....
UNTIL Found OR Index > MaxIndex

IF Found THEN
    OUTPUT "Value found at position ", Index
ELSE
    OUTPUT .....
ENDIF

```

[4]

(b) (i) State the necessary condition for a binary search.

.....
 [1]

(ii) Describe how to perform a binary search.

.....

 [4]

- (iii) Explain how the performance of a binary search varies according to the number of values in the array.

.....

.....

..... [1]

- (c) Compare the performance of the algorithms for a binary search and a linear search using Big O notation for order of time complexity.

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..... [3]

- 9 State the reasons for including exception handling routines when writing a program.

Include an example of an exception in your answer.

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..... [4]

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.....

.....

.....

..... [2]

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.....

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.....

.....

.....

..... [2]

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.....

.....

.....

.....

.....

.....

[3]

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05 type(persian, domestic).
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12
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18

19 [2]

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```

returns

```
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```

Write the result returned by the goal:

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```

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.....
..... [1]

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.....
.....
.....
..... [2]

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Acrylic
.....
Constructor()

Wool
.....
Constructor()

Mix
Percentage: INTEGER
Constructor() YarnInfo()

[5]

(b) Describe what is meant by the terms **properties**, **methods** and **inheritance**.

Properties

.....

.....

.....

Methods

.....

.....

.....

Inheritance

.....

.....

.....

[6]

- 3 (a) State what is meant by the term **enumerated data type**.

.....
 [1]

- (b) State what is meant by the term **pointer data type**.

.....
 [1]

- (c) The months of the year are: January, February, March, April, May, June, July, August, September, October, November and December.

Write the **pseudocode** statement to define the data type `Quarter1`, to hold the names of the first three months of a year.

.....

 [2]

- (d) The composite data type `Pet` is used to store data about the various pets of a group of students. It uses these fields:

Field name	Data type
<code>PetName</code>	String
<code>AnimalType</code>	String
<code>PetAge</code>	Integer
<code>PetGender</code>	Char
<code>OwnerName</code>	String

- (i) Write the **pseudocode** statement to set up a variable for one record of the composite data type `Pet`.

.....
 [1]

- (ii) Write **pseudocode** to store the details of the following pet, in the variable you set up in part (d)(i).

PetName	AnimalType	PetAge	PetGender	OwnerName
Tibbles	Cat	8	M	Jasmine Smith

.....

.....

.....

.....

.....

..... [3]

- 4 Draw **one** line to connect each stage of compilation to its **most appropriate** description.

Stage of compilation

Description

Lexical analysis

Syntax analysis

Code generation

Optimisation

minimising a program's execution time and memory requirement

converting an intermediate representation of source code into an executable form

converting a sequence of characters into a sequence of tokens

directly executing instructions written in a scripting language

using parsing algorithms to interpret the meaning of a sequence of tokens

[4]

11 (a) Define these Object-Oriented Programming (OOP) terms:

Instance

.....

Inheritance

.....

Polymorphism

.....

[3]

(b) In OOP, a class contains attributes and methods.

Complete the pseudocode for the class `Car` to enable objects to be created. The class needs to include:

- string attributes to store the make, model, body type and fuel type
- an integer attribute to store the number of cars of that type built.

The attributes must be available only through the methods of the class.

CLASS

PRIVATE Make : STRING

PRIVATE

.....

.....

.....

PUBLIC PROCEDURE New(CarMake : STRING,,

.....)

Make ←

Model ←

BodyType ← CarBodyType

Fuel ← ""

NumberBuilt ← 0

ENDPROCEDURE

GetFuel()

GetNumberBuilt()

.....

[5]

- 12 (a)** The array `Names[0:99]` is in alphabetical order.

Complete this pseudocode binary search algorithm.

`Lower ← 0`

.....
`Mid ← 0`

`Exit ← FALSE`

`OUTPUT "Enter the name to be found "`

`INPUT Target`

`REPEAT`

..... `THEN`

`OUTPUT Target, " does not exist"`

`Exit ← TRUE`

`ENDIF`

`Mid ← Lower + (Upper - Lower + 1) DIV 2`

`IF Names[Mid] < Target THEN`

`Lower ← .....`

`ENDIF`

`IF Names[Mid] > Target THEN`

.....
`ENDIF`

..... `THEN`

`OUTPUT Target, " was found at location ", Mid`

`Exit ← TRUE`

`ENDIF`

.....

[6]

- (b)** Big O notation is used to classify efficiency of algorithms.

The Big O notation for time complexity in a binary search is $O(\log n)$.

- (i)** State the Big O notation for time complexity of a linear search.

..... [1]

- (ii)** Describe the meaning of $O(\log n)$ as it applies to a binary search algorithm.

.....

.....

.....

..... [2]

4 A program to manage regular flight details at an airport requires some user-defined data types.

- (a) Write **pseudocode** statements to declare the enumerated data type `Aircraft` to hold data about the types of aircraft used for a flight.

These types of aircraft are: C300, C350, D242, E757, X380.

.....

.....

.....

..... [2]

- (b) Write **pseudocode** statements to declare the composite data type `Flight` to hold data about flights to a specific destination. These include:

- flight number, which could be any combination of letters and numbers
- destination
- date of departure
- type of aircraft used.

Use the enumerated data type you created in **part (a)**.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (c) (i) Write the **pseudocode** statement to set up a variable for one record of the composite data type `Flight`.

.....

..... [1]

- (ii) Write **pseudocode** to store the details of the following flight in the variable you set up in **part (c)(i)**.

Field	Data
flight number	XA782
destination	Cambridge
date of departure	12/12/2022
type of aircraft used	C350

Use the field names you created in **part (b)**.

.....

.....

.....

.....

.....

..... [3]

- 5** Describe what is meant by a **virtual machine**.
Include in your answer **two** benefits and **two** drawbacks of using a virtual machine.

Description

.....

.....

.....

Benefit 1

.....

Benefit 2

.....

Drawback 1

.....

Drawback 2

.....

[6]

10 (a) Define these Object-Oriented Programming (OOP) terms:

Encapsulation

.....

Getter

.....

Setter

.....

[3]

(b) A school has a program written using OOP to maintain its staff and student records.

The object `SubstituteTeacher` allows the details of the school's substitute teachers to be stored. This includes their full name, telephone number and whether or not they are in school today. For example:

SubName	Sarah Jones
Telephone	01223658721
InSchool	TRUE

Complete the diagram for the object `SubstituteTeacher`, including appropriate properties and their getters and setters.

SubstituteTeacher
SubName : STRING
.....
InSchool : BOOLEAN
.....
SetTelephone(Tel : STRING)
.....
GetSubName ()
.....
.....

[3]

- 11 A simplified linked list is used to store the names of flowers in alphabetical order. It is implemented using two 1D arrays:

- `Flower` stores the names of the flowers.
- `NextPointer` stores the pointer to the next flower name in the list.

`HeadPointer` indicates the index of the first flower name in the linked list.

`HeadPointer` 6

When the end of the linked list is reached, the next pointer has the value of 0.

The following table shows the initial content of the arrays.

Index	Flower	NextPointer
1	Rose	7
2	Marigold	1
3	Foxglove	10
4	Iris	9
5	Daisy	3
6	Dahlia	5
7	Saxifrage	0
8	Lupin	2
9	Lily	8
10	Hydrangea	4

- (a) Several flower names have been deleted from the linked list. These are crossed out in the following table.

Complete the table to show the new values of `HeadPointer` and `NextPointer` to keep the remaining flower names in alphabetical order.

`HeadPointer`

Index	Flower	NextPointer
1	Rose	
2	Marigold	
3	Foxglove	
4	Iris	
5	Daisy	
6	Dahlia	
7	Saxifrage	
8	Lupin	
9	Lily	
10	Hydrangea	

(b) Complete the pseudocode algorithm so that it achieves the following when applied to the arrays:

- The flower name is input.
- The linked list is searched, in order, for the flower name.
- If the flower name is found, an appropriate message is output to indicate it has been found.
- If the flower name is not found, an appropriate message is output to indicate it has not been found.
- The algorithm terminates when the next pointer value is 0.

Pointer $\leftarrow$ HeadPointer

Found $\leftarrow$ 0

OUTPUT "Enter a flower name "

.....

.....

IF Flower[Pointer] = FlowerName THEN

Found $\leftarrow$ Pointer

Pointer $\leftarrow$ 0

ELSE

.....

ENDIF

ENDWHILE

.....

OUTPUT Flower[Found], " is found"

ELSE

.....

ENDIF

[5]

(c) Explain how you could improve the simplified linked list structure.

.....

.....

.....

..... [2]

11 (a) Define these Object-Oriented Programming (OOP) terms:

Instance

.....

Inheritance

.....

Polymorphism

.....

[3]

(b) In OOP, a class contains attributes and methods.

Complete the pseudocode for the class `Car` to enable objects to be created. The class needs to include:

- string attributes to store the make, model, body type and fuel type
- an integer attribute to store the number of cars of that type built.

The attributes must be available only through the methods of the class.

CLASS

PRIVATE Make : STRING

PRIVATE

.....

.....

.....

PUBLIC PROCEDURE New(CarMake : STRING,,

.....)

Make ←

Model ←

BodyType ← CarBodyType

Fuel ← ""

NumberBuilt ← 0

ENDPROCEDURE

GetFuel()

GetNumberBuilt()

.....

[5]

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`Lower ← 0`

.....
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`Exit ← FALSE`

`OUTPUT "Enter the name to be found "`

`INPUT Target`

`REPEAT`

..... `THEN`

`OUTPUT Target, " does not exist"`

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`ENDIF`

`Mid ← Lower + (Upper - Lower + 1) DIV 2`

`IF Names[Mid] < Target THEN`

`Lower ← .....`

`ENDIF`

`IF Names[Mid] > Target THEN`

.....
`ENDIF`

..... `THEN`

`OUTPUT Target, " was found at location ", Mid`

`Exit ← TRUE`

`ENDIF`

..... [6]

- (b)** Big O notation is used to classify efficiency of algorithms.

The Big O notation for time complexity in a binary search is $O(\log n)$.

- (i)** State the Big O notation for time complexity of a linear search.

..... [1]

- (ii)** Describe the meaning of $O(\log n)$ as it applies to a binary search algorithm.

.....

.....

.....

..... [2]

- 10 The pseudocode algorithm shown copies an active accounts text file `ActiveFile.txt` to an archive accounts text file `ArchiveFile.txt`, one line at a time. Any blank lines found in the active accounts text file are replaced with the words "Account not present" in the archive accounts text file.

Complete this file-handling pseudocode.

```
DECLARE Account : STRING
```

```
.....

OPENFILE "ArchiveFile.txt" FOR WRITE

WHILE NOT .....
    READFILE "ActiveFile.txt", Account
    IF Account = "" THEN

        WRITEFILE "ArchiveFile.txt", " ....."
    ELSE

        WRITEFILE "ArchiveFile.txt", .....
    ENDIF
ENDWHILE

.....

CLOSEFILE "ArchiveFile.txt"
```

[5]

- 11 Pseudocode is to be written to implement a queue Abstract Data Type (ADT) with items of the string data type. This will be implemented using the information in the table.

Identifier	Data type	Description
FrontPointer	INTEGER	points to the start of the queue
RearPointer	INTEGER	points to the end of the queue
Length	INTEGER	the current size of the queue
Queue	STRING	1D array to implement the queue

A constant, with identifier `MaxSize`, limits the size of the queue to 60 items.

- (a) Write the pseudocode to declare `MaxSize`, `FrontPointer`, `RearPointer`, `Length` and `Queue`.

```
.....

.....

.....

.....

.....

.....

.....
```

[3]

- (b) Complete the following pseudocode for the function `Dequeue` to remove the front item from the queue.

```

FUNCTION Dequeue RETURNS STRING
  DECLARE Item : STRING

  ..... > 0 THEN

    Item ← .....

    .....
    IF Length = 0 THEN
      CALL Initialise // reset the pointers
    ELSE
      IF FrontPointer > MaxSize THEN

        ..... ← 1
      ENDIF
    ENDIF
  ELSE
    OUTPUT "The print queue was empty - error!"
    Item ← ""
  ENDIF
  RETURN Item
ENDFUNCTION

```

[4]

- (c) Explain how a new element can be added to the queue if it is implemented using two stacks.

.....

.....

.....

.....

.....

..... [4]

- 12 (a) Describe what is meant by recursion.

.....

.....

.....

..... [2]

- (b) A Fibonacci sequence is a series of numbers formed by adding together the two preceding numbers, for example:

0, 1, 1, 2, ...

This function calculates and returns values in the Fibonacci sequence and uses recursion.

```
FUNCTION Fib(Number : INTEGER) RETURNS INTEGER
  IF Number <= 1 THEN
    Result ← Number
  ELSE
    Result ← Fib(Number - 1) + Fib(Number - 2)
  ENDIF
  RETURN Result
ENDFUNCTION
```

Complete the trace table for the function when it is called as `Fib(5)`.

Call number	Function call	Number	Result	Return value
		5		

[5]

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Cambridge Assessment International Education is part of Cambridge Assessment. Cambridge Assessment is the brand name of the University of Cambridge Local Examinations Syndicate (UCLES), which is a department of the University of Cambridge.

- 4 Draw **one** line from each Object-Oriented Programming (OOP) term to its **most appropriate** description.

OOP term	Description
Encapsulation	methods used to return the value of a property
Getters	the process of putting data and methods together as a single unit
Polymorphism	methods used to update the value of a property
Setters	allows methods to be redefined for derived classes
	enables the defining of a new class that inherits from a parent class

[4]

- 5 (a) Encryption is used to scramble data to make it meaningless if intercepted.

Describe the purpose of quantum cryptography.

.....

.....

.....

..... [2]

- (b) Explain the differences between symmetric and asymmetric cryptography when encrypting and decrypting data.

.....

.....

.....

.....

.....

..... [3]

- 6 (a) Write **pseudocode** statements to declare the composite data type, `TAppointments`, to hold data about patients for a dental clinic. It will include for each patient:

- name (first name and last name)
- date of birth
- telephone number
- date of last appointment
- date of next appointment
- all treatments are complete (yes or no).

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) This pseudocode algorithm reads dental records stored in a random file using the user-defined data type `TAppointments` and prints the contents of the file, one record at a time.

Complete this file handling pseudocode:

```
DECLARE DentalRecord : ARRAY[1:250] OF TAppointments
DECLARE DentalFile : STRING
DECLARE Count : INTEGER
DentalFile ← "DentalFile.dat"
OUTPUT "The file ", DentalFile, " contains these records:"
```

```
OPENFILE .....
```

```
..... ← 1
```

```
REPEAT
```

```
    SEEK DentalFile, Count
```

```
    .....
    OUTPUT DentalRecord[Count]
```

```
    Count ← Count + 1
```

```
.....(DentalFile)
```

```
.....
```

[5]

- 11 (a) The pseudocode shown represents a queue Abstract Data Type (ADT) with procedures for initialisation and to add new items. It is incomplete.

```

CONSTANT MaxLength = 50
DECLARE FrontPointer : INTEGER
DECLARE RearPointer : INTEGER
DECLARE Length : INTEGER
DECLARE Queue : ARRAY[0 : MaxLength - 1] OF STRING

// initialisation of queue
PROCEDURE Initialise
    FrontPointer ← -1

    .....

    ..... ← 0
ENDPROCEDURE

// adding a new item to the queue
PROCEDURE Enqueue(NewItem : STRING)

    IF ..... THEN

        RearPointer ← .....
        IF RearPointer > MaxLength - 1 THEN
            RearPointer ← 0
        ENDIF

        .....
        Length ← Length + 1
    ENDIF
ENDPROCEDURE

```

- (i) Study the pseudocode and insert the identifiers to complete this table.

Identifier	Data type	Description
	STRING	An array to store the contents of the queue.
	INTEGER	Points to the last item of the queue.
	INTEGER	Indicates the number of items in the queue.
	INTEGER	Points to the first item of the queue.

[2]

- (ii) Complete the given pseudocode.

[5]

- (b) Explain the reasons why a queue ADT works better than a stack ADT in organising print jobs.

.....

.....

.....

.....

.....

..... [3]

- 10 The pseudocode algorithm shown copies an active accounts text file `ActiveFile.txt` to an archive accounts text file `ArchiveFile.txt`, one line at a time. Any blank lines found in the active accounts text file are replaced with the words "Account not present" in the archive accounts text file.

Complete this file-handling pseudocode.

```
DECLARE Account : STRING
```

```
.....

OPENFILE "ArchiveFile.txt" FOR WRITE

WHILE NOT .....
    READFILE "ActiveFile.txt", Account
    IF Account = "" THEN

        WRITEFILE "ArchiveFile.txt", " ....."
    ELSE

        WRITEFILE "ArchiveFile.txt", .....
    ENDIF
ENDWHILE

.....

CLOSEFILE "ArchiveFile.txt"
```

[5]

- 11 Pseudocode is to be written to implement a queue Abstract Data Type (ADT) with items of the string data type. This will be implemented using the information in the table.

Identifier	Data type	Description
FrontPointer	INTEGER	points to the start of the queue
RearPointer	INTEGER	points to the end of the queue
Length	INTEGER	the current size of the queue
Queue	STRING	1D array to implement the queue

A constant, with identifier `MaxSize`, limits the size of the queue to 60 items.

- (a) Write the pseudocode to declare `MaxSize`, `FrontPointer`, `RearPointer`, `Length` and `Queue`.

```
.....

.....

.....

.....

.....

.....

.....
```

[3]

- (b) Complete the following pseudocode for the function `Dequeue` to remove the front item from the queue.

```

FUNCTION Dequeue RETURNS STRING
  DECLARE Item : STRING

  ..... > 0 THEN

    Item ← .....

    .....
    IF Length = 0 THEN
      CALL Initialise // reset the pointers
    ELSE
      IF FrontPointer > MaxSize THEN

        ..... ← 1
      ENDIF
    ENDIF
  ELSE
    OUTPUT "The print queue was empty - error!"
    Item ← ""
  ENDIF
  RETURN Item
ENDFUNCTION

```

[4]

- (c) Explain how a new element can be added to the queue if it is implemented using two stacks.

.....

.....

.....

.....

.....

..... [4]

- 12 (a) Describe what is meant by recursion.

.....

.....

.....

..... [2]

- (b) A Fibonacci sequence is a series of numbers formed by adding together the two preceding numbers, for example:

0, 1, 1, 2, ...

This function calculates and returns values in the Fibonacci sequence and uses recursion.

```
FUNCTION Fib(Number : INTEGER) RETURNS INTEGER
  IF Number <= 1 THEN
    Result ← Number
  ELSE
    Result ← Fib(Number - 1) + Fib(Number - 2)
  ENDIF
  RETURN Result
ENDFUNCTION
```

Complete the trace table for the function when it is called as `Fib(5)`.

Call number	Function call	Number	Result	Return value
		5		

[5]

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- 2 (a) Draw **one** line to connect **each** protocol to its most appropriate use.

Protocol	Use
HTTP	to provide peer-to-peer file sharing
BitTorrent	when retrieving email messages from a mail server over a TCP/IP connection
SMTP	when transmitting hypertext documents
IMAP	to map MAC addresses onto IP addresses
	when sending email messages towards the intended destination

[4]

- (b) Outline the purpose of the **Link layer** in the TCP/IP protocol suite.

.....

.....

.....

..... [2]

- 3 Describe what is meant by **enumerated** and **pointer** data types.

Enumerated

.....

.....

.....

Pointer

.....

.....

.....

[4]

- 8 (a) A pseudocode algorithm finds a customer account record in a random file and outputs it. The records are stored using the user-defined data type `TAccount`.

```

TYPE TAccount
    DECLARE AccountNumber : INTEGER
    DECLARE LastName : STRING
    DECLARE FirstName : STRING
    DECLARE Address : STRING
    DECLARE ContactNumber : STRING
ENDTYPE

```

Complete the file handling pseudocode.

The function `Hash()` takes the customer account number as a parameter, calculates and returns the hash value.

```

DECLARE Customer : TAccount
DECLARE Location : INTEGER
DECLARE AccountFile : STRING

..... ← "AccountRecords.dat"

..... AccountFile .....

OUTPUT "Please enter an account number"
INPUT Customer.AccountNumber

Location ← Hash(.....)

SEEK ....., Location

..... AccountFile, .....

OUTPUT Customer          // output customer record
CLOSEFILE AccountFile

```

[5]

- (b) Define the term **exception handling**.

.....
 [1]

- (c) State **two** possible causes of an exception.

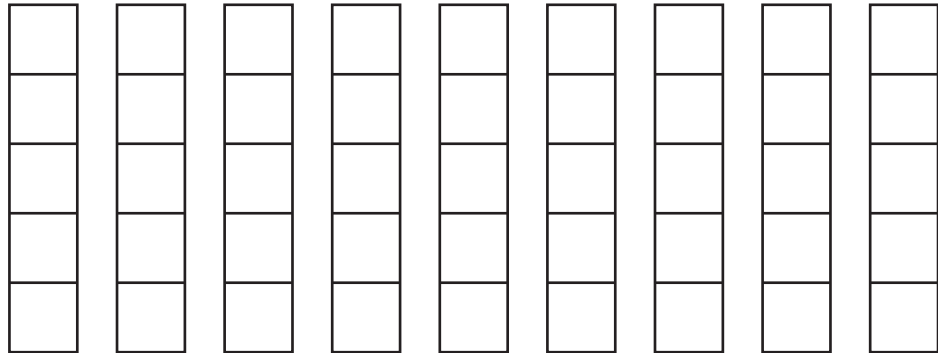
.....
 [2]

- 9 (a) (i) Write the infix expression for this Reverse Polish Notation (RPN) expression:

5 2 - 5 4 + * 9 /

.....
 [2]

- (ii) Show how the contents of the following stack will change as the RPN expression in part (a)(i) is evaluated.



[4]

- (b) Explain how a stack can be used to evaluate RPN expressions.

.....

 [3]

10 A stack is to be set up using the information in the table.

Identifier	Data type	Description
BasePointer	INTEGER	points to the bottom of the stack
TopPointer	INTEGER	points to the top of the stack
Stack	REAL	1D array to implement the stack

A constant, with identifier `Capacity`, limits the size of the stack to 25 items.

(a) Write the **pseudocode** for the required declarations.

.....

 [3]

(b) Complete the pseudocode function `Pop()` to pop an item from `Stack`.

```
// popping an item from the stack

FUNCTION Pop() .....
  DECLARE Item : REAL
  Item ← 0

  ..... BasePointer THEN

    Item ← .....

    TopPointer ← .....
  ELSE
    OUTPUT "The stack is empty - error"
  ENDIF

  .....

ENDFUNCTION
```

[5]

(c) Compare and contrast the queue and stack Abstract Data Types (ADT).

.....

.....

.....

.....

.....

..... [2]

- 11 A **declarative** programming language is used to represent subjects that students can choose to study.

Students must choose two subjects.

```

01 subject(mathematics).
02 subject(physics).
03 subject(chemistry).
04 subject(computer_science).
05 subject(geography).
06 subject(history).
07 subject(english).
08 subject(biology).
09 student(tomaz).
10 student(josephine).
11 student(elspeth).
12 student(nico).
13 student(teresa).
14 student(pietre).
15 choice1(tomaz, mathematics).
16 choice1(teresa, chemistry).
17 choice1(pietre, mathematics).
18 choice1(nico, mathematics).
19 choice1(elspeth, chemistry).
20 choice2(tomaz, computer_science).
21 choice2(nico, geography).

```

These clauses have the meanings:

Clause	Meaning
01	Mathematics is a subject.
09	Tomaz is a student.
15	Tomaz has chosen mathematics as his first choice.
20	Tomaz has chosen computer science as his second choice.

- (a) Anthony is a student who would like to study history and geography.

Write additional clauses to represent this information.

22
 23
 24 [3]

- (b) Using the variable x , the goal:

`choice1(X, chemistry)`

returns

`X = teresa, elspeth`

Write the result returned by the goal:

`choice1(X, mathematics)`

`X = .....` [1]

- (c) Students must choose two different subjects such that:

N may choose S, if N is a student and S is a subject and N has not chosen S as the first choice.

Write this as a rule.

`may_choose_subject(N, S)`

IF

 [4]

- 9 (a) A stack Abstract Data Type (ADT) is to be implemented using pseudocode, with procedures to initialise it and to push new items onto the stack.

A 1D array `Stack` stores the contents of the stack.

- (i) Study the pseudocode in **part (a)(ii)** and complete the table of identifiers by writing the missing data types and descriptions.

Identifier	Data type	Description
BasePointer		
TopPointer		
Stack	REAL	

[2]

- (ii) Complete the pseudocode.

```

CONSTANT MaxSize = 40
DECLARE BasePointer : INTEGER
DECLARE TopPointer : INTEGER
DECLARE Stack : ARRAY[1:40] OF REAL

// initialisation of stack
PROCEDURE Initialise()

    ..... ← 1

    ..... ← 0
ENDPROCEDURE

// push an item onto the stack
PROCEDURE Push(NewItem : REAL)

    ..... MaxSize THEN

    .....

    Stack[TopPointer] ← .....
ENDIF
ENDPROCEDURE

```

[5]

(b) Justify the use of a linked list instead of an array to implement a stack.

.....

.....

.....

..... [2]

(c) Explain how a compiler makes use of a stack when translating recursive programming code.

.....

.....

.....

.....

.....

.....

.....

..... [4]

10 Describe the features of the SIMD and MISD computer architectures.

SIMD

.....

.....

.....

MISD

.....

.....

..... [4]

- 11** A **declarative** programming language is used to represent some facts about people and their hobbies.

```

01 hobby(music) .
02 hobby(caving) .
03 hobby(climbing) .
04 hobby(camping) .
05 hobby(baking) .
06 hobby(travelling) .
07 person(toby) .
08 person(natasha) .
09 person(fatima) .
10 person(joseph) .
11 person(elijah) .
12 person(nina) .
13 enjoys(natasha, travelling) .
14 enjoys(toby, climbing) .
15 enjoys(nina, climbing) .
16 enjoys(elijah, camping) .
17 enjoys(fatima, baking) .
18 enjoys(joseph, camping) .
19 dislikes(toby, caving) .

```

These clauses have the meanings:

Clause	Meaning
01	Music is a hobby
07	Toby is a person
13	Natasha enjoys travelling
19	Toby dislikes caving

- (a)** Carlos is a person who enjoys the hobby of cycling but does not like music.

Write additional clauses to represent this information.

```

20 .....
21 .....
22 .....
23 .....

```

[4]

(b) Using the variable *P*, the goal:

`enjoys(P, camping)`

returns

`P = elijah, joseph`

Write the result returned by the goal:

`enjoys(P, climbing)`

`P = .....` [1]

(c) N is a person who might enjoy H if H is a hobby and N does not dislike H.

Write this as a rule.

`might_enjoy(N, H)`

IF

 [4]

12 (a) Describe, with an example, what is meant by an **exception**.

.....

 [2]

- (b) A pseudocode algorithm searches for a customer record in a random file `AccountRecord.dat`. A user inputs the name of the customer.

The records are stored using the user-defined data type `TAccount`.

```
TYPE TAccount
    DECLARE AccountNumber : INTEGER
    DECLARE Name : STRING
    DECLARE Address : STRING
    DECLARE Telephone : STRING
ENDTYPE
```

If the record is found, it is output, otherwise an error message is displayed.

Complete the file handling pseudocode.

```
DECLARE Customer : TAccount
DECLARE Location : INTEGER
DECLARE MaxSize : INTEGER
DECLARE FoundFlag : BOOLEAN
DECLARE SearchCustomer : STRING
MaxSize ← 1000

OPENFILE .....
Location ← 1

..... ← FALSE
OUTPUT "Enter the customer's name"

.....

..... AND Location <= MaxSize

..... "AccountRecord.dat", .....
GETRECORD "AccountRecord.dat", Customer
IF SearchCustomer = Customer.Name THEN
    OUTPUT "Customer found: "
    OUTPUT Customer           // output customer record
    FoundFlag ← TRUE
ENDIF
Location ← Location + 1
ENDWHILE
IF NOT FoundFlag THEN

    OUTPUT "....."
ENDIF
```

- 8 (a) A pseudocode algorithm finds a customer account record in a random file and outputs it. The records are stored using the user-defined data type `TAccount`.

```

TYPE TAccount
    DECLARE AccountNumber : INTEGER
    DECLARE LastName : STRING
    DECLARE FirstName : STRING
    DECLARE Address : STRING
    DECLARE ContactNumber : STRING
ENDTYPE

```

Complete the file handling pseudocode.

The function `Hash()` takes the customer account number as a parameter, calculates and returns the hash value.

```

DECLARE Customer : TAccount
DECLARE Location : INTEGER
DECLARE AccountFile : STRING

..... ← "AccountRecords.dat"

..... AccountFile .....

OUTPUT "Please enter an account number"
INPUT Customer.AccountNumber

Location ← Hash(.....)

SEEK ....., Location

..... AccountFile, .....

OUTPUT Customer          // output customer record
CLOSEFILE AccountFile

```

[5]

- (b) Define the term **exception handling**.

.....
 [1]

- (c) State **two** possible causes of an exception.

.....
 [2]

10 A stack is to be set up using the information in the table.

Identifier	Data type	Description
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Stack	REAL	1D array to implement the stack

A constant, with identifier `Capacity`, limits the size of the stack to 25 items.

(a) Write the **pseudocode** for the required declarations.

.....

.....

.....

.....

.....

..... [3]

(b) Complete the pseudocode function `Pop()` to pop an item from `Stack`.

```
// popping an item from the stack

FUNCTION Pop() .....
    DECLARE Item : REAL
    Item ← 0

    ..... BasePointer THEN

        Item ← .....

        TopPointer ← .....
    ELSE
        OUTPUT "The stack is empty - error"
    ENDIF

    .....

ENDFUNCTION
```

[5]

(c) Compare and contrast the queue and stack Abstract Data Types (ADT).

.....

.....

.....

.....

.....

..... [2]

- 11 A **declarative** programming language is used to represent subjects that students can choose to study.

Students must choose two subjects.

```

01 subject(mathematics).
02 subject(physics).
03 subject(chemistry).
04 subject(computer_science).
05 subject(geography).
06 subject(history).
07 subject(english).
08 subject(biology).
09 student(tomaz).
10 student(josephine).
11 student(elspeth).
12 student(nico).
13 student(teresa).
14 student(pietre).
15 choice1(tomaz, mathematics).
16 choice1(teresa, chemistry).
17 choice1(pietre, mathematics).
18 choice1(nico, mathematics).
19 choice1(elspeth, chemistry).
20 choice2(tomaz, computer_science).
21 choice2(nico, geography).

```

These clauses have the meanings:

Clause	Meaning
01	Mathematics is a subject.
09	Tomaz is a student.
15	Tomaz has chosen mathematics as his first choice.
20	Tomaz has chosen computer science as his second choice.

- (a) Anthony is a student who would like to study history and geography.

Write additional clauses to represent this information.

22
 23
 24 [3]

- (b) Using the variable X , the goal:

`choice1(X, chemistry)`

returns

`X = teresa, elspeth`

Write the result returned by the goal:

`choice1(X, mathematics)`

`X = .....` [1]

- (c) Students must choose two different subjects such that:

N may choose S , if N is a student and S is a subject and N has not chosen S as the first choice.

Write this as a rule.

`may_choose_subject(N, S)`

IF

 [4]

- 8 A **declarative** programming language is used to represent the features that are available and the features that are unavailable on different body styles of a car.

```

01 feature(sunroof).
02 feature(automatic_tailgate).
03 feature(heated_seats).
04 feature(extra_seats).
05 feature(reversing_camera).
06 feature(dashboard_camera).
07 feature(air_conditioning).
08 feature(heated_windscreen).
09 feature(satnav).
10 bodystyle(saloon).
11 bodystyle(hatchback).
12 bodystyle(estate).
13 bodystyle(minivan).
14 bodystyle(convertible).
15 available(sunroof, hatchback).
16 available(sunroof, minivan).
17 available(reversing_camera, hatchback).
18 available(extra_seats, minivan).
19 available(reversing_camera, saloon).
20 unavailable(sunroof, convertible).
21 unavailable(automatic_tailgate, saloon).
22 unavailable(extra_seats, hatchback).

```

These clauses have the meanings:

Clause	Meaning
01	Sunroof is a feature.
10	Saloon is a body style.
15	Sunroof is available on a hatchback.
20	Sunroof is unavailable on a convertible.

- (a) Sliding doors is a feature that is available on a minivan but unavailable on a hatchback.

Write additional clauses to represent this information.

23

24

25

[3]

(b) Using the variable `Options`, the goal:

```
available(Options, saloon)
```

returns

```
Options = reversing_camera
```

Write the result returned by the goal:

```
available(Options, hatchback)
```

`Options = .....` [1]

(c) `F` may be available for `B` if `F` is a feature and `B` is a body style and `F` is **not** unavailable for that body style.

Write this as a rule:

```
may_choose_option(F, B)
```

IF [4]

9 Explain what is meant by **Deep Learning** in relation to Artificial Intelligence (AI).

..... [3]

- 10 (a) State a condition that must be true for an array to be searchable for a binary search.

.....
 [1]

- (b) Complete the given pseudocode to find an item in a 1D array `Names` of type `STRING` using a binary search.

```

DECLARE Names : ARRAY[1:100000] OF STRING
DECLARE TopOfList : INTEGER
DECLARE EndOfList : INTEGER
DECLARE CurrentItem : INTEGER
DECLARE ToFind : STRING
DECLARE Found : BOOLEAN
DECLARE NotInList : BOOLEAN
TopOfList ← 1
EndOfList ← 100000

OUTPUT "Which name do you wish to find? "
INPUT ToFind

.....

NotInList ← FALSE

WHILE ..... AND .....
    CurrentItem ← (TopOfList + EndOfList) DIV 2

    IF ..... THEN
        Found ← TRUE
    ELSE
        IF TopOfList >= EndOfList THEN
            .....
        ELSE
            IF ToFind > Names[CurrentItem] THEN
                .....
            ELSE
                EndOfList ← CurrentItem - 1
            ENDIF
        ENDIF
    ENDIF
ENDWHILE
IF Found = TRUE THEN
    OUTPUT "Item found at position ", CurrentItem, " in array"
ELSE
    OUTPUT "Item not in array"
ENDIF
  
```

[5]

- (c) Describe the performance of a binary search in relation to the number of data items in the array being searched. Refer to Big O notation in your answer.

.....

.....

.....

..... [2]

- 11 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

- (a) State **two** features of RISC processors.

.....

.....

.....

..... [2]

- (b) Outline the process of interrupt handling as it could be applied to RISC or CISC processors.

.....

.....

.....

.....

.....

.....

..... [3]

- (c) Explain how pipelining affects interrupt handling for RISC processors.

.....

.....

.....

.....

.....

..... [3]

- 8 (a) Complete the pseudocode to find an item in a 1D array `Widgets` of type `STRING`, using a **linear** search.

```

DECLARE Widgets : ARRAY[1:50000] OF STRING
DECLARE TopOfList : INTEGER
DECLARE EndOfList : INTEGER
DECLARE Count : INTEGER
DECLARE ToFind : STRING
DECLARE Found : BOOLEAN
DECLARE NotInList : BOOLEAN

TopOfList ← 1
EndOfList ← 50000
OUTPUT "Enter the name of the item you wish to find "
INPUT ToFind

.....

NotInList ← FALSE
Count ← TopOfList

WHILE ..... AND .....

    IF ..... THEN
        Found ← TRUE
    ENDIF
    Count ← Count + 1

    IF ..... THEN
        NotInList ← TRUE
    ENDIF
ENDWHILE

IF Found = TRUE THEN
    OUTPUT "Item found at position ", Count - 1, " in array"
ELSE
    OUTPUT "Item not in array"
ENDIF

```

[4]

- (b)** Compare the methods used by the linear and binary search algorithms to find an item in an array. Refer to Big O notation in your answer.

..... [4

[4]

- 10** A **declarative** programming language is used to allow clients to choose daily activities at the beach.

```

01 activity(paddleboarding).
02 activity(sailing).
03 activity(rowing).
04 activity(kayaking).
05 activity(jetskiing).
06 client(stevie).
07 client(antonio).
08 client(henry).
09 client(eliza).
10 client(rebeka).
11 client(danny).
12 client(erik).
13 client(simone).
14 client(petra).
15 client(frankie).
16 choice(petra, rowing).
17 choice(frankie, sailing).
18 choice(erik, sailing).
19 choice(eliza, rowing).
20 choice(stevie, jetskiing).
21 choice(henry, sailing).
22 done(henry, jetskiing).
23 done(rebeka, jetskiing).
24 done(antonio, kayaking).

```

These clauses have the meanings:

Clause	Meaning
01	Paddle boarding is an activity.
06	Stevie is a client.
16	Petra has chosen rowing.
22	Henry has already done jet skiing.

- (a)** Jane is a client who would like to choose the activity surfing and she has already done sailing.

Write additional clauses to represent this information.

```

25 .....
26 .....
27 .....
28 .....

```

[4]

(b) Using the variable `List`, the goal:

```
choice(List, rowing)
```

returns

```
List = petra, eliza
```

Write the result returned by the goal:

```
choice(List, sailing)
```

List = [1]

(c) C is a client who would like to choose A if A is an activity and C has **not** already done A.

Write this as a rule:

```
may_choose_activity(C, A)
```

IF [4]

11 Explain what is meant by **Reinforcement Learning** in relation to Artificial Intelligence.

..... [3]

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- 8 A **declarative** programming language is used to represent the features that are available and the features that are unavailable on different body styles of a car.

```

01 feature(sunroof) .
02 feature(automatic_tailgate) .
03 feature(heated_seats) .
04 feature(extra_seats) .
05 feature(reversing_camera) .
06 feature(dashboard_camera) .
07 feature(air_conditioning) .
08 feature(heated_windscreen) .
09 feature(satnav) .
10 bodystyle(saloon) .
11 bodystyle(hatchback) .
12 bodystyle(estate) .
13 bodystyle(minivan) .
14 bodystyle(convertible) .
15 available(sunroof, hatchback) .
16 available(sunroof, minivan) .
17 available(reversing_camera, hatchback) .
18 available(extra_seats, minivan) .
19 available(reversing_camera, saloon) .
20 unavailable(sunroof, convertible) .
21 unavailable(automatic_tailgate, saloon) .
22 unavailable(extra_seats, hatchback) .

```

These clauses have the meanings:

Clause	Meaning
01	Sunroof is a feature.
10	Saloon is a body style.
15	Sunroof is available on a hatchback.
20	Sunroof is unavailable on a convertible.

- (a) Sliding doors is a feature that is available on a minivan but unavailable on a hatchback.

Write additional clauses to represent this information.

23

24

25

[3]





(b) Using the variable `Options`, the goal:

`available(Options, saloon)`

returns

`Options = reversing_camera`

Write the result returned by the goal:

`available(Options, hatchback)`

`Options = .....` [1]

(c) `F` may be available for `B` if `F` is a feature and `B` is a body style and `F` is **not** unavailable for that body style.

Write this as a rule:

`may_choose_option(F, B)`

IF
.....
.....
..... [4]

9 Explain what is meant by **Deep Learning** in relation to Artificial Intelligence (AI).

.....
.....
.....
.....
..... [3]





10 (a) State a condition that must be true for an array to be searchable for a binary search.

.....
 [1]

(b) Complete the given pseudocode to find an item in a 1D array `Names` of type `STRING` using a binary search.

```

DECLARE Names : ARRAY[1:100000] OF STRING
DECLARE TopOfList : INTEGER
DECLARE EndOfList : INTEGER
DECLARE CurrentItem : INTEGER
DECLARE ToFind : STRING
DECLARE Found : BOOLEAN
DECLARE NotInList : BOOLEAN
TopOfList ← 1
EndOfList ← 100000

```

```

OUTPUT "Which name do you wish to find? "
INPUT ToFind

```

.....

```

NotInList ← FALSE

```

```

WHILE ..... AND .....
    CurrentItem ← (TopOfList + EndOfList) DIV 2

```

```

    IF ..... THEN
        Found ← TRUE

```

```

    ELSE
        IF TopOfList >= EndOfList THEN

```

.....

```

        ELSE
            IF ToFind > Names[CurrentItem] THEN

```

.....

```

            ELSE
                EndOfList ← CurrentItem - 1
            ENDIF

```

```

        ENDIF

```

```

    ENDIF

```

```

ENDWHILE

```

```

IF Found = TRUE THEN

```

```

    OUTPUT "Item found at position ", CurrentItem, " in array"

```

```

ELSE

```

```

    OUTPUT "Item not in array"

```

```

ENDIF

```

[5]





- (c) Describe the performance of a binary search in relation to the number of data items in the array being searched. Refer to Big O notation in your answer.

.....

.....

.....

..... [2]

- 11 Reduced Instruction Set Computers (RISC) and Complex Instruction Set Computers (CISC) are two types of processor.

- (a) State **two** features of RISC processors.

.....

.....

.....

..... [2]

- (b) Outline the process of interrupt handling as it could be applied to RISC or CISC processors.

.....

.....

.....

.....

.....

.....

..... [3]

- (c) Explain how pipelining affects interrupt handling for RISC processors.

.....

.....

.....

.....

.....

.....

..... [3]





- 9 A veterinary surgery wants to create a class for individual pets. Some of the attributes required in the class are listed in the table.

Attribute	Data type	Description
PetID	STRING	unique ID assigned at registration
PetType	STRING	type of pet assigned at registration
OwnerTelephone	STRING	telephone number of owner assigned at registration
DateRegistered	DATE	date of registration

- (a) State **one** reason why the attributes would be declared as `PRIVATE`.

.....
..... [1]

- (b) Complete the class diagram for `Pet`, to include:

- an attribute and data type for the name of the pet
- an attribute and data type for the name of the owner
- a method to create a `Pet` object and set attributes at the time of registration
- a method to assign a pet ID
- a method to assign the date of registration
- a method to return the pet name
- a method to return the owner's telephone number.

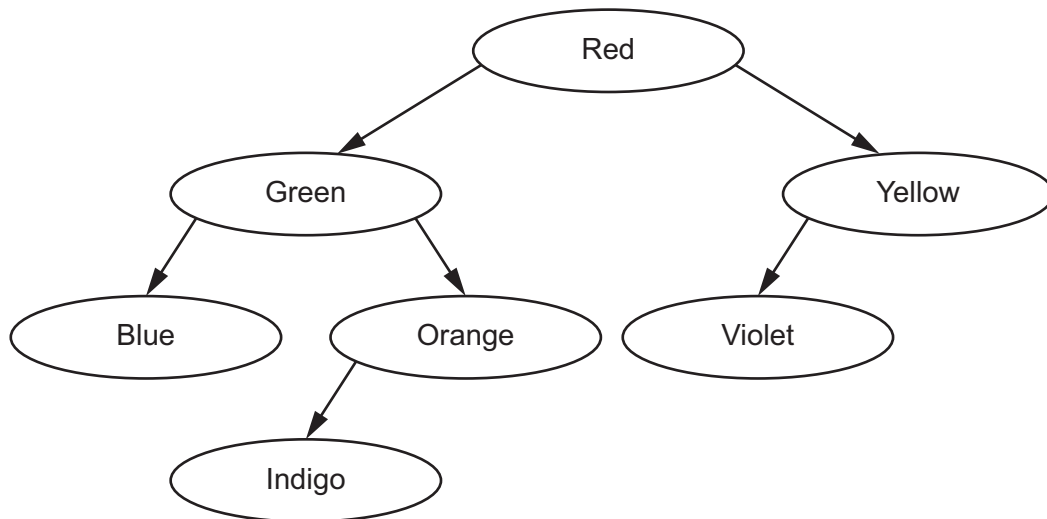
Pet	
PetID	: STRING
PetType	: STRING
OwnerTelephone	: STRING
DateRegistered	: DATE
.....	:
.....	:
.....	
.....	
.....	

[5]





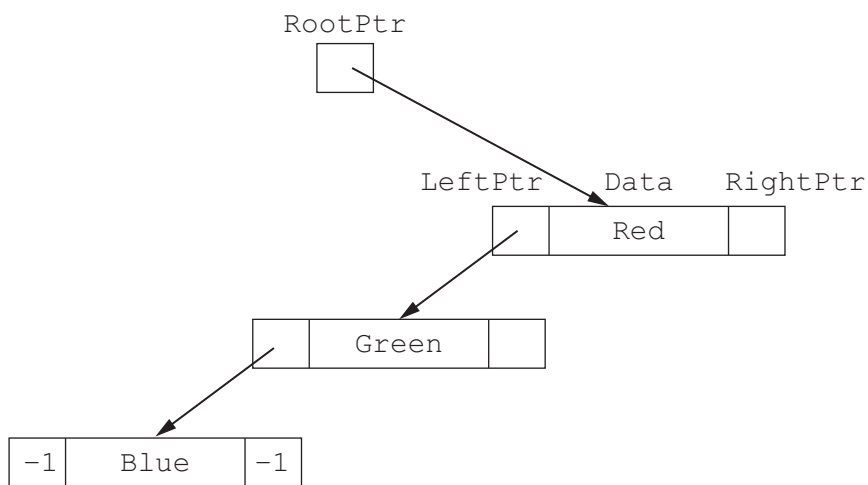
11 The following diagram shows an ordered binary tree.



- (a) A linked list of nodes is used to store the data. Each node consists of a left pointer, the data and a right pointer.

–1 is used to represent a null pointer.

Complete this linked list to represent the given binary tree.





(b) A user-defined record structure is used to store the nodes of the linked list in part (a).

Complete the diagram, using your answer for part (a).

RootPtr	Index	LeftPtr	Data	RightPtr
0	0		Red	
	1		Green	
	2		Yellow	
	3		Blue	
	4		Orange	
	5		Indigo	
FreePtr	6		Violet	
	7			

[4]

(c) The linked list in part (a) is implemented using a 1D array of records. Each record contains a left pointer, data and a right pointer.

The following pseudocode represents a function that searches for an element in the array of records `BinTree`. It returns the index of the record if the element is found, or it returns a null pointer if the element is **not** found.

Complete the pseudocode for the function.

```

FUNCTION SearchTree(Item : STRING) .....

    NowPtr ← .....

    WHILE NowPtr <> -1

        IF ..... THEN

            NowPtr ← BinTree[NowPtr].LeftPtr

        ELSE

            IF BinTree[NowPtr].Data < Item THEN

                .....

            ELSE

                RETURN NowPtr

            ENDIF

        ENDIF

    ENDWHILE

    RETURN NowPtr

ENDFUNCTION
    
```

[4]





3 (a) Describe the user-defined data type **record**.

.....

.....

.....

.....

.....

..... [3]

(b) A programmer defines a record, `Order`, to store the following data:

- account number
- order number
- order price
- order date.

Write **pseudocode** statements to define this record.

.....

.....

.....

.....

.....

.....

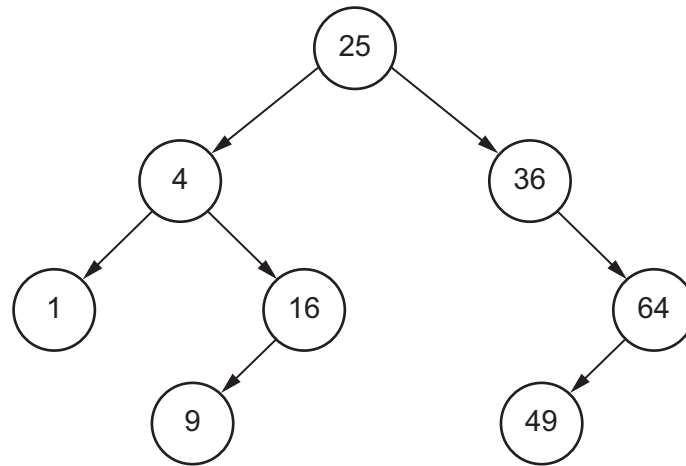
.....

..... [4]





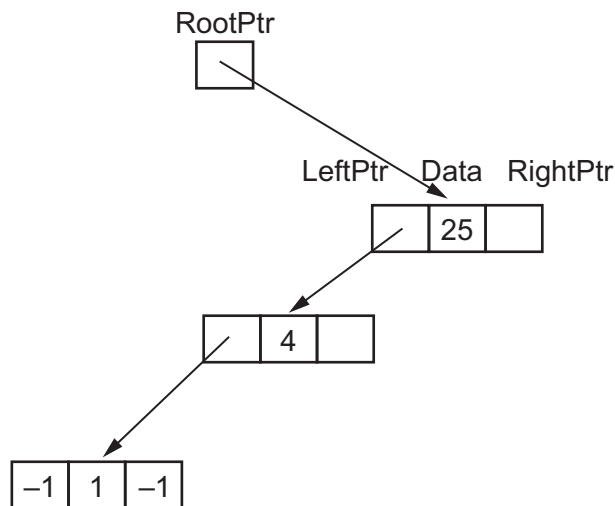
11 This binary tree shows an ordered list of integers.



(a) A linked list of nodes is used to store the data. Each node consists of a left pointer, the data and a right pointer.

–1 is used to represent a null pointer.

Complete this linked list to represent the given binary tree organisation.



[4]



- (b) A 2D array is used to store the nodes of the linked list in part (a).

Complete the diagram using your answer for part (a).

RootPtr	Index	LeftPtr	Data	RightPtr
0	0		25	
	1		4	
	2		36	
	3		1	
	4		16	
	5		64	
FreePtr	6		9	
	7		49	
	8			

[4]

- (c) The linked list in part (a) is implemented using a 1D array of records. Each record contains a left pointer, data and a right pointer.

The following pseudocode represents a function that searches for an element in the array of records `LinkList`. It returns the index of the record if the element is found, or it returns a null pointer if the element is not found.

Complete the pseudocode for the function.

```

FUNCTION SearchList(Item : INTEGER).....
    NullPtr ← -1

    ..... ← RootPtr
    WHILE NowPtr <> NullPtr
        IF LinkList[NowPtr].Data < Item THEN
            NowPtr ← LinkList[NowPtr].RightPtr
        ELSE
            IF ..... THEN

                NowPtr ← .....
            ELSE
                RETURN NowPtr
            ENDIF
        ENDIF
    ENDWHILE
    RETURN NullPtr
ENDFUNCTION

```

[4]



- 9 A veterinary surgery wants to create a class for individual pets. Some of the attributes required in the class are listed in the table.

Attribute	Data type	Description
PetID	STRING	unique ID assigned at registration
PetType	STRING	type of pet assigned at registration
OwnerTelephone	STRING	telephone number of owner assigned at registration
DateRegistered	DATE	date of registration

- (a) State **one** reason why the attributes would be declared as `PRIVATE`.

.....
 [1]

- (b) Complete the class diagram for `Pet`, to include:

- an attribute and data type for the name of the pet
- an attribute and data type for the name of the owner
- a method to create a `Pet` object and set attributes at the time of registration
- a method to assign a pet ID
- a method to assign the date of registration
- a method to return the pet name
- a method to return the owner's telephone number.

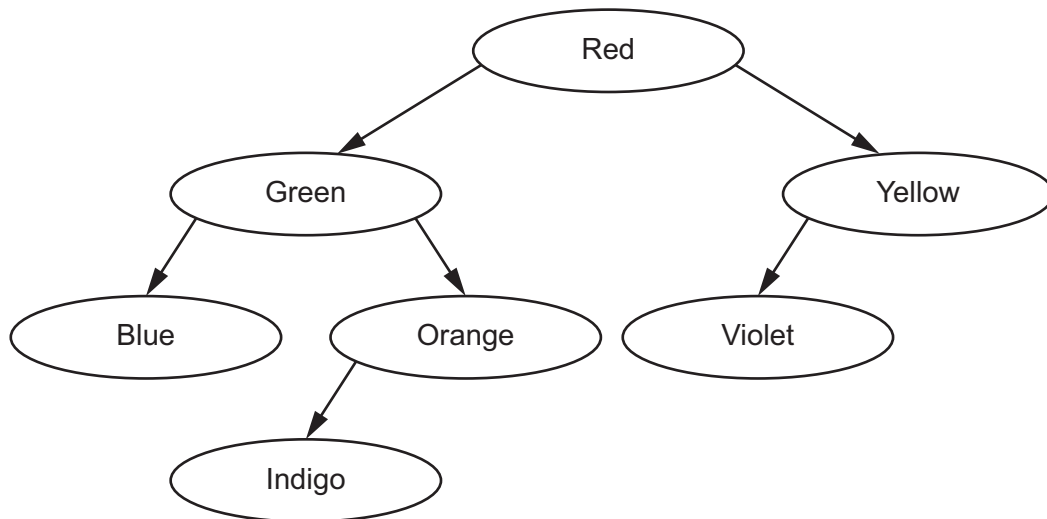
Pet	
PetID	: STRING
PetType	: STRING
OwnerTelephone	: STRING
DateRegistered	: DATE
.....	:
.....	:
.....	
.....	
.....	

[5]





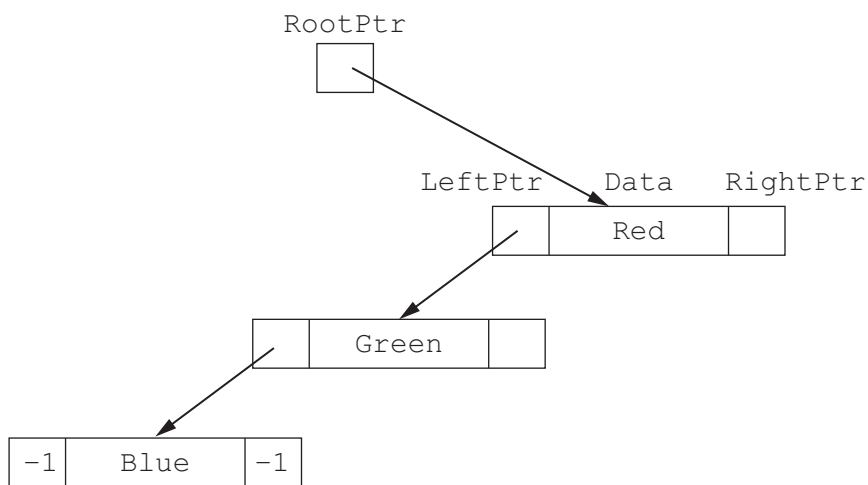
11 The following diagram shows an ordered binary tree.



- (a) A linked list of nodes is used to store the data. Each node consists of a left pointer, the data and a right pointer.

–1 is used to represent a null pointer.

Complete this linked list to represent the given binary tree.





(b) A user-defined record structure is used to store the nodes of the linked list in part (a).

Complete the diagram, using your answer for part (a).

RootPtr	Index	LeftPtr	Data	RightPtr
0	0		Red	
	1		Green	
	2		Yellow	
	3		Blue	
	4		Orange	
	5		Indigo	
FreePtr	6		Violet	
	7			

[4]

(c) The linked list in part (a) is implemented using a 1D array of records. Each record contains a left pointer, data and a right pointer.

The following pseudocode represents a function that searches for an element in the array of records `BinTree`. It returns the index of the record if the element is found, or it returns a null pointer if the element is **not** found.

Complete the pseudocode for the function.

```

FUNCTION SearchTree(Item : STRING) .....

    NowPtr ← .....

    WHILE NowPtr <> -1

        IF ..... THEN

            NowPtr ← BinTree[NowPtr].LeftPtr

        ELSE

            IF BinTree[NowPtr].Data < Item THEN

                .....

            ELSE

                RETURN NowPtr

            ENDIF

        ENDIF

    ENDWHILE

    RETURN NowPtr

ENDFUNCTION
    
```

[4]





- 1 A programmer is writing a program to manage bookings for a small taxi company. The programmer requires some user-defined data types.

- (a) Write a **pseudocode** statement to declare the enumerated data type, `Vehicle`, to hold the identity code of each of the company's taxis:

M100, M230, T101, T102, T120, T150

.....
 [2]

- (b) Write **pseudocode** statements to declare the composite data type, `Booking`, to hold data about taxi bookings. The data required includes:

- booking number (any combination of letters and numbers)
- destination
- client name
- client telephone number
- date of departure
- address for pick-up
- the identity code of the taxi used.

Use the most appropriate data type in each case, including the enumerated data type from part (a).

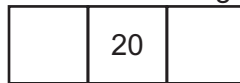
.....

 [4]



- 4 (a) A linked list of nodes is used to store an ordered list of integers. Each node consists of the data, a left pointer and a right pointer, for example:

Left pointer Data Right pointer

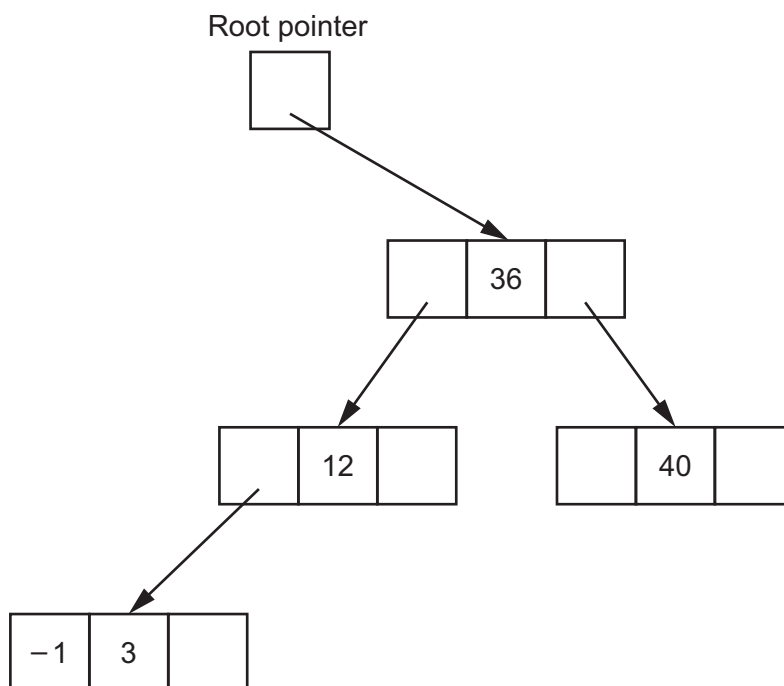


The linked list will be organised as a binary tree.

-1 is used to represent a null pointer.

Complete the binary tree, including null pointers, to show how the data will be organised after the following integers have been added:

6, 15, 41, 66



[4]

- (b) Describe what is meant by recursion.

.....

.....

.....

..... [2]





- (c) A binary tree is a suitable Abstract Data Type (ADT) that a designer can implement using recursive algorithms.

Identify **one other** ADT that a designer can implement using recursive algorithms.

..... [1]

- 5 This truth table represents a logic circuit.

INPUT				OUTPUT
A	B	C	D	Z
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

- (a) Write the Boolean logic expression that corresponds to the given truth table as the sum-of-products.

Z =

..... [2]



- DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

DO NOT WRITE IN THIS MARGIN

- attribute and data type for the identification of the patient
- attribute and data type for the identification of the doctor
- methods to assign date seen, treatments given and medications prescribed
- method to return the date seen and the attributes for the patient and the doctor.

[5]

- (b) (i) Identify the object-oriented programming (OOP) feature whose function includes restricting external access to the data.

..... [1]

- (ii) Describe what is meant by the OOP feature **inheritance**.

.....

 [2]

- 12 The pseudocode algorithm checks whether a location in a stock file `StockList.dat` is empty or not. The location is given by the user. If the location is empty, a suitable message is displayed, otherwise the item stored at that location is displayed.

Complete this file-handling pseudocode algorithm.

```
DECLARE Location : INTEGER
DECLARE Item : STRING
DECLARE Continue : BOOLEAN
DECLARE Answer : CHAR
Continue ← TRUE
```

```
OPENFILE .....
WHILE Continue
  OUTPUT "Enter a location between 1 and 500: "
  INPUT Location
```

```
  GETRECORD .....
  IF Item = "" THEN
    OUTPUT "This record is missing."
  ELSE
    OUTPUT "The item in stock is ", .....
  ENDIF
  OUTPUT "Another location (Y or N)?"
  INPUT Answer
  IF Answer <> 'Y' THEN
    Continue ← FALSE
  ENDIF
ENDWHILE
```

```
.....
OUTPUT "End of program"
```

[5]





1 Data types can be defined using pseudocode.

The composite record data type, `Departure`, is used to represent flights from Cambridge Airport and is defined in pseudocode as:

```
TYPE Departure
  DECLARE FlightNumber : STRING
  DECLARE Destination : STRING
  DECLARE FlightDate : DATE
  DECLARE Gate : STRING
  DECLARE Airline : STRING
ENDTYPE
```

A variable, `Flight1`, is declared in pseudocode as:

```
DECLARE Flight1 : Departure
```

(a) Write **pseudocode** to store the following details to `Flight1`:

Field	Data
FlightNumber	SB2789
Destination	Dublin
FlightDate	30/07/2025
Gate	N03
Airline	Cambridge Airways

.....

.....

.....

.....

.....

..... [3]

(b) The data type for `Gate` is changed to an enumerated data type, `GateID`.

(i) Write a **pseudocode** statement to declare `GateID` to hold the identity codes for the airport gates:

N01, N02, N03, W01, W02, W03, W04

.....

..... [2]

(ii) Write the new **pseudocode** statement required to replace the declaration of `Gate` in `Departure`.

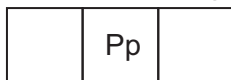
..... [1]





- 11 (a) A linked list of nodes is used to store an ordered list of strings. Each node consists of the data, a left pointer and a right pointer.

Left pointer Data Right pointer

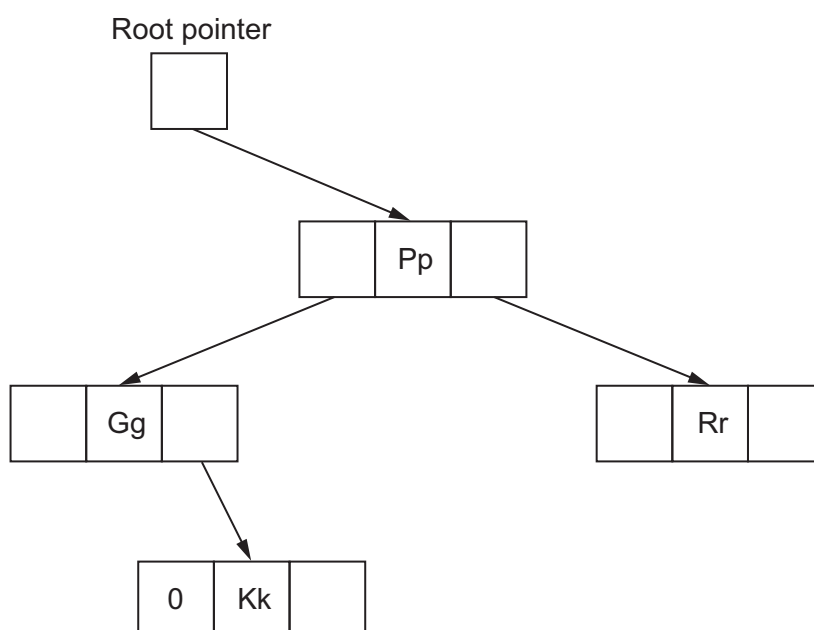


The linked list will be organised as a binary tree.

0 is used to represent a null pointer.

Complete the binary tree, including null pointers, to show how the data will be organised after the following strings have been added:

Aa, Mm, Ss, Xx



[4]





(b) A binary tree can be used to implement recursion.

Identify **one** feature of an algorithm that makes it beneficial to use recursion.
Give **one** example of an application that could use a recursive algorithm.

Feature

.....

Example

[2]





- 12 (a) A medical clinic uses objects of the class `Patient` to assign a priority and a doctor to a patient. Some of the attributes required in the class are listed in the table.

Attribute	Data type	Description
PatientID	STRING	Unique identifier of the patient
Name	STRING	Patient's full name, surname first
DoctorID	STRING	ID of doctor administering treatment

Treatment is prioritised with a numeric scale of 1 to 5.

Complete the class diagram for `Patient`, to include:

- attribute and data type for the date of birth
- attribute and data type for the priority
- methods to assign the patient ID, priority and doctor ID
- methods to return the patient ID, patient date of birth and the priority.

Patient	
PatientID	: STRING
Name	: STRING
.....	:
.....	:
DoctorID	: STRING
.....	
SetName(FullName : STRING)	
SetDateOfBirth(DOB : DATE)	
.....	
.....	
.....	
GetName()	
.....	
.....	
GetDoctorID()	

[5]



- (b) (i) Identify the object-oriented programming (OOP) term described as 'an occurrence of an object'.

..... [1]

- (ii) Describe what is meant by the OOP term **polymorphism**.

.....

 [2]

- 13 The pseudocode algorithm below uses random file access to copy 50 records from a live file `CurrentResults.dat` to a stored file `StoredResults.dat` one record at a time. It uses the user-defined type `StudentResult`.

```
TYPE StudentResult
  DECLARE LastName : STRING
  DECLARE FirstName : STRING
  DECLARE ExamGrade : STRING
ENDTYPE
```

If any grades are missing in `CurrentResults.dat`, the text "Missing grade" is added to the `ExamGrade` field in `StoredResults.dat`

Complete this file handling pseudocode algorithm.

```
DECLARE Grade : StudentResult
DECLARE Position : INTEGER
```

```
.....
OPENFILE "StoredResults.dat" FOR RANDOM
```

```
.....
  SEEK "CurrentResults.dat", Position
  GETRECORD "CurrentResults.dat", Grade
  IF Grade.ExamGrade = "" THEN
```

```
  .....
  ENDIF
```

```
  .....
  .....
  NEXT Position
```

```
CLOSEFILE "CurrentResults.dat"
CLOSEFILE "StoredResults.dat"
```

[5]





1 A programmer is writing a program to manage a video library. They require a user-defined data type.

(a) Write **pseudocode** statements to declare the composite data type `VideoLibrary` to hold data about each video in the collection. This data includes:

- identity code (any combination of letters and numbers)
- title
- year released
- date purchased
- format (for example DVD, Blu-ray, 4K, MP4)
- running time (minutes)

Use the most appropriate data type in each case.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) Identify **one** field in `VideoLibrary` that could be an efficient enumerated data type and give a reason for your choice.

Field

.....

Reason

..... [2]





- | OOP term | Description |
|----------|---|
| | A method that accesses the value of a property. |
| | A method that changes the value of a property. |
| Object |
.....
..... |
| Method |
.....
..... |

6 The management and scheduling of processes are tasks carried out by an operating system.

- [2]

- Function
-
-
-
-
-
-
- Benefit

- 11 The pseudocode algorithm below allows a user to input a new stock item. The random file is searched for the next empty location in the file and the new item is inserted there. A suitable message is displayed if the file is full.

Complete this pseudocode.

```

DECLARE Location : INTEGER
DECLARE NewStock : STRING
DECLARE CurrentStock : STRING
DECLARE Stored : BOOLEAN
DECLARE Max : INTEGER
Max ← 100000
Stored ← FALSE
Location ← 1

```

```

.....
OUTPUT "Enter the new item you wish to store: "
INPUT NewStock
WHILE NOT Stored AND Location <= Max

```

```

.....
GETRECORD "StockList.dat", .....
IF CurrentStock = "" THEN
    ..... "StockList.dat", NewStock
    Stored ← TRUE
ELSE
    Location ← Location + 1
ENDIF
ENDWHILE

..... THEN
    OUTPUT "The new item has not been stored as the file was full."
ENDIF
CLOSEFILE "StockList.dat"

```

[5]





12 (a) An array is an Abstract Data Type (ADT).

Identify **two** other ADTs.

1

2 [1]

(b) A 1D array `dataArray` holds up to 1000 elements of type integer and needs to be sorted in ascending order.

Write the **pseudocode** for an insertion sort to sort the array into ascending order.

Use the identifiers from the table in your algorithm.

You do **not** need to declare any arrays or variables for this algorithm. You may assume this has already been done.

Identifier	Data type	Description
Index	INTEGER	counter for outer loop
Position	INTEGER	counter for inner loop – insertion position
dataArray	INTEGER	1D array to store up to 1000 integers
Value	INTEGER	value to insert

The first line has been written for you.

FOR Index $\leftarrow$ 2 to 1000

.....

 [4]





(c) Describe **two** ways in which the performance of a sort routine is affected by the data to be sorted.

1

.....

2

.....

[2]





13 The recursive procedure `Delete()` is defined as follows:

```
PROCEDURE Delete(Index, Target)
  IF Numbers[Index] > 0 THEN
    IF Numbers[Index] >= Target THEN
      Numbers[Index] ← Numbers[Index + 1]
    ENDIF
    Index ← Index + 1
    CALL Delete(Index, Target)
  ENDIF
ENDPROCEDURE
```

An array `Numbers` is used to store a sorted data set of non-zero positive integers. Unused cells contain zero.

The contents of the array at the start of the algorithm are:

Numbers									
[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
2	3	7	11	15	17	19	23	0	0

Complete the trace table for the algorithm for the procedure call:

`CALL Delete(1, 15)`

Index	Target	Numbers									
		[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]	[10]
		2	3	7	11	15	17	19	23	0	0

[4]

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- 1 The composite record data type, *ClubMember*, is defined in pseudocode as:

```

TYPE ClubMember
  DECLARE Code : INTEGER
  DECLARE LastName : STRING
  DECLARE FirstName : STRING
  DECLARE Telephone : STRING
  DECLARE JoinDate : DATE
  DECLARE Fees : REAL
  DECLARE FeesPaid : BOOLEAN
ENDTYPE

```

- (a) (i) Write the **pseudocode** statement to set up a variable for one record of the composite data type, *ClubMember*.

.....
 [1]

- (ii) Write the **pseudocode** statements to assign the following values to the variable set up in part (a)(i):

- 984632 to Code
- TRUE to FeesPaid

.....

 [2]

- (b) An enumerated data type, *Activity*, is required, so that a new field, *Choice*, can be added to the composite data type, *ClubMember*, to allow members to choose an activity.

- (i) Write the **pseudocode** statement for the type declaration of *Activity* to hold the names of the available activities:

Badminton, Football, Golf, Snooker, Swimming, Tennis.

.....

 [2]

- (ii) Write the **new pseudocode** statement required to update the declaration of *Choice* in the definition of *ClubMember*.

.....
 [1]





- 9 (a) A stack has been implemented using pseudocode to store a maximum of 100 string items using the global variables in the following table:

Identifier	Data type	Description	Initialisation value
Base	INTEGER	pointer for the bottom of the stack	0
Top	INTEGER	pointer for the top of the stack	-1
StackArray	STRING	1D array to implement the stack	[0:99]
Max	INTEGER	maximum number of items in the stack	100

The value of `Top` is incremented each time a data item is added to the stack and decremented every time a data item is removed.

- (i) Complete the **pseudocode** for the function to remove a data item from the stack.

```

FUNCTION Pop() .....
    DECLARE DataItem : STRING
    DataItem ← ""

    IF ..... THEN

        DataItem ← .....

        Top ← .....
    ELSE
        DataItem ← "You cannot remove data; the stack is empty"
    ENDIF

    .....
ENDFUNCTION
    
```

[5]

- (ii) Write the **pseudocode** to output the data item removed from the stack with an appropriate message.

.....

..... [1]





(b) A stack is used to implement recursion.

State the **three** essential features of recursion.

- 1
-
- 2
-
- 3
-

[3]

10 Explain what is meant by **exception handling**.
Include an example of a possible cause of an exception in your answer.

Explanation

.....

.....

.....

Example

[3]



- 1 The composite data type, *Car*, is defined in pseudocode as:

```

TYPE Car
  DECLARE RegNumber : STRING
  DECLARE Make : STRING
  DECLARE Model : STRING
  DECLARE BodyStyle : STRING
  DECLARE Colour : STRING
  DECLARE IntoStock : DATE
  DECLARE Price : REAL
ENDTYPE

```

- (a) (i) Write the **pseudocode** statement to set up a variable for one record of the composite data type, *Car*.

.....
 [1]

- (ii) Write the **pseudocode** statements to assign the following values to the variable set up in part (a)(i):

- "Blue" to Colour
- 21/10/2025 to IntoStock

.....

 [2]

- (b) The data type for *BodyStyle* is changed to an enumerated type, *Body*.

- (i) Write the **pseudocode** statement for the type declaration of *Body* to hold the names of the available choices:

Convertible, Hatchback, Saloon, SUV

.....

 [2]

- (ii) Write the new **pseudocode** statement required to update the declaration of *BodyStyle* in the definition of *Car*.

.....
 [1]





- 12 (a) A stack has been implemented using pseudocode to store a maximum of 100 string items using the global variables in the following table:

Identifier	Data type	Description	Initialisation value
Base	INTEGER	pointer for the bottom of the stack	0
Top	INTEGER	pointer for the top of the stack	-1
StackArray	STRING	1D array to implement the stack	[0:99]
Max	INTEGER	maximum number of items in the stack	100

The value of `Top` is incremented each time a data item is added to the stack and decremented each time a data item is removed. If the stack is full, an appropriate error message is output.

- (i) Complete the **pseudocode** for the procedure to add a data item onto the stack.

```

PROCEDURE Push (.....)
  IF Top < Max - 1 THEN

    Top ← .....

    ..... ← NewData
  ELSE

    OUTPUT .....
  ENDIF
ENDPROCEDURE

```

[4]

- (ii) Write **pseudocode** to input a new data item and add it to the stack using `Push()`.

```

.....
.....
.....
.....

```

[2]

- (b) Explain the reasons why a stack is used when a recursive algorithm is executed.

```

.....
.....
.....
.....
.....
.....

```

[3]



1 (a) An enumerated data type, `Spectrum`, is required to hold the names of colours.

(i) Write the **pseudocode** statement for the type declaration of `Spectrum` to hold the names of the colours available:

Red, Orange, Yellow, Green, Blue, Indigo, Violet.

.....
 [2]

(ii) Identify **two** characteristics of the list of values given in an enumerated data type declaration.

1

 2
 [2]

(b) `ColourData` is a composite data type to store definitions of colours.

Write **pseudocode** statements to declare `ColourData` to hold the following fields:

Field	Example data
ColourCode	XYZ12345
Colour	Red
Wavelength	650
Frequency	4.62
PrimaryColour	Yes

Use the most appropriate data type in each case, including the enumerated data type `Spectrum` from part a(i).

.....

 [4]

